Machine-Learning Model Implementation

for LSATISFY

1. Model Development

After preprocessing and feature selection, the final dataset contained **39,999 observations and 40 predictor variables**. The target variable was LSATISFY, a 4-level life-satisfaction variable ranging from 1 to 4.

The data were divided into 80% training data (31,999 observations) and 20% testing data (8,000 observations) using random_state = 42. The target distribution was approximately 45% for category 1, 49% for category 2, 4% for category 3, and 1% for category 4.

Four conventional regression models were implemented:

- XGBoost
- Random Forest
- LightGBM
- Linear Regression

Tree-based models were trained using the original predictor values, while the predictors were standardized for Linear Regression.

2. Hyperparameter Tuning and Validation

Hyperparameters for XGBoost, Random Forest, and LightGBM were optimized using 5-fold cross-validation, with Mean Absolute Error (MAE) as the optimization criterion.

3. Model Performance

Model	Test MAE	Test RMSE	Test R^2
XGBoost	0.416	0.509	**0.337**
Random Forest	0.417	0.513	0.326
LightGBM	**0.415**	0.510	0.333
Linear Regression	0.432	0.518	0.312

Among the conventional models, LightGBM achieved the lowest MAE (0.415), while XGBoost achieved the highest R^2 (0.337). The relatively similar cross-validation and test errors indicate that the models generalized reasonably well to unseen data.

4. Summary

Overall, the nonlinear models performed better than Linear Regression, indicating that the relationship between the BRFSS predictors and life satisfaction is not entirely linear. However, the R^2 values of approximately 0.31–0.34 indicate that a substantial portion of the variability in life satisfaction remains unexplained by the selected predictors.